

Name:

Matriculation number:

MST-Technologies and Processes

SS 2019

Duration: 120 min.

Please note

- write your name and matriculation number into the caption of every sheet
- use and write only on the front of the sheet
- use a separate sheet for every task

Cheating

- any attempt of cheating results in immediate failure of the exam
- switch off your cell phones and store them in your backpack – a cell phone on the table will be treated as an attempt to cheating

Withdrawal or absence

- you may withdraw until the moment the tasks are revealed
- please see:
https://www.tf.uni-freiburg.de/en/studies-and-teaching/a-to-z-study-faq/de-registration-of-exams?set_language=en
- you find the relevant forms here:
<https://www.tf.uni-freiburg.de/de/studium-lehre/a-bis-z-studium/dokumente/WithdrawalfromExamsduetoIllnessMasterofScience.pdf>

Necessary Points:

60 of 120 in total

Points:

Result:



1. Processes**22**

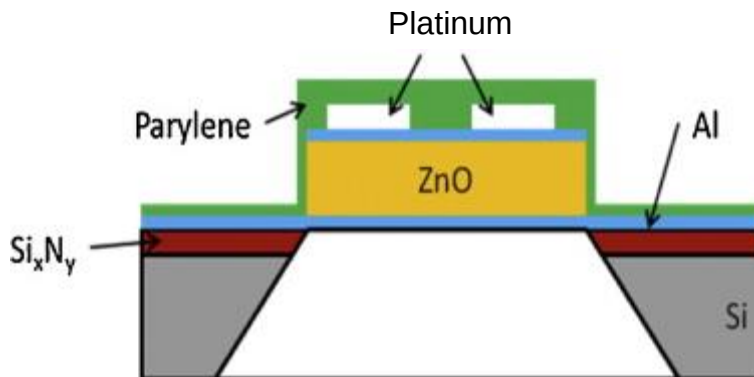
1.1 Processes

6

name 2 processes from each of the following process classes:
transformative, subtractive, additive

1.2 Sketch the manufacturing steps required to fabricate the following structure (a biosensor)

16

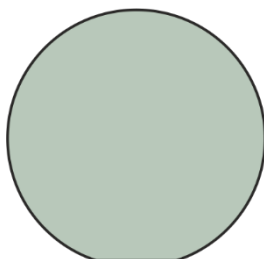
**2. Cleanroom technology****16**

- 2.1 Name four general recommendations for working in the cleanroom 4
- 2.2 What are common ISO classes used for white areas in a cleanroom for MEMS and for microelectronics? 4
- 2.3 What are the two common cleanroom setups? 2
- 2.4 What are the four filtering concepts used in filter-fan units? 4
- 2.5 What is the Neutra system and what is its purpose? 2

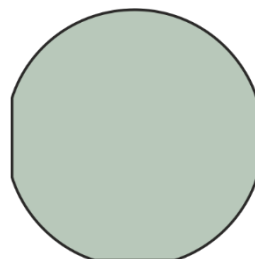
3. Semiconductor materials**32**

- 3.1 Name and sketch 4 different Bravais lattices 4
- 3.2 Name two common semiconductor materials 2
- 3.3 What is their most outstanding physical properties (for which they would be most likely chosen in an application) 4
- 3.4 State the wafer type for the given flats (assume p doping on both cases) 4

a)



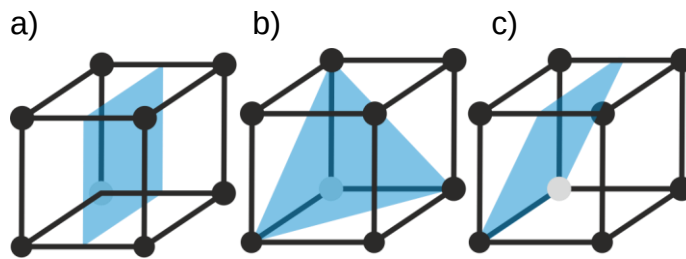
b)



3.5 Indicate the planes in the following unit cells

9





- | | | |
|-----|---|---|
| 3.6 | Explain and sketch 0-, 1-, 2- and 3-dimensional defects in crystals | 4 |
| 3.7 | Explain and sketch the steps of the Czochralski process | 5 |

4. Photolithography 30

- | | | |
|-----|---|---|
| 4.1 | Sketch and explain the process workflow for generating a chromium mask using a mask-less lithography | 6 |
| 4.2 | Why are anti-reflective coatings used in photolithography? | 2 |
| 4.3 | Which thickness is required for an anti-reflective coating made from a material with $n=1.5$ for a wavelength of 600 nm | 4 |
| 4.4 | Name three common substrate materials for photomasks | 3 |
| 4.5 | What is diffraction? | 4 |
| 4.6 | What is the Rayleigh criterion? State the equation for the minimal feature resolution according to the Rayleigh criterion | 3 |
| 4.7 | Sketch the lithographic setup in proximity lithography | 6 |
| 4.8 | Name the two relevant classes of photoresists | 2 |

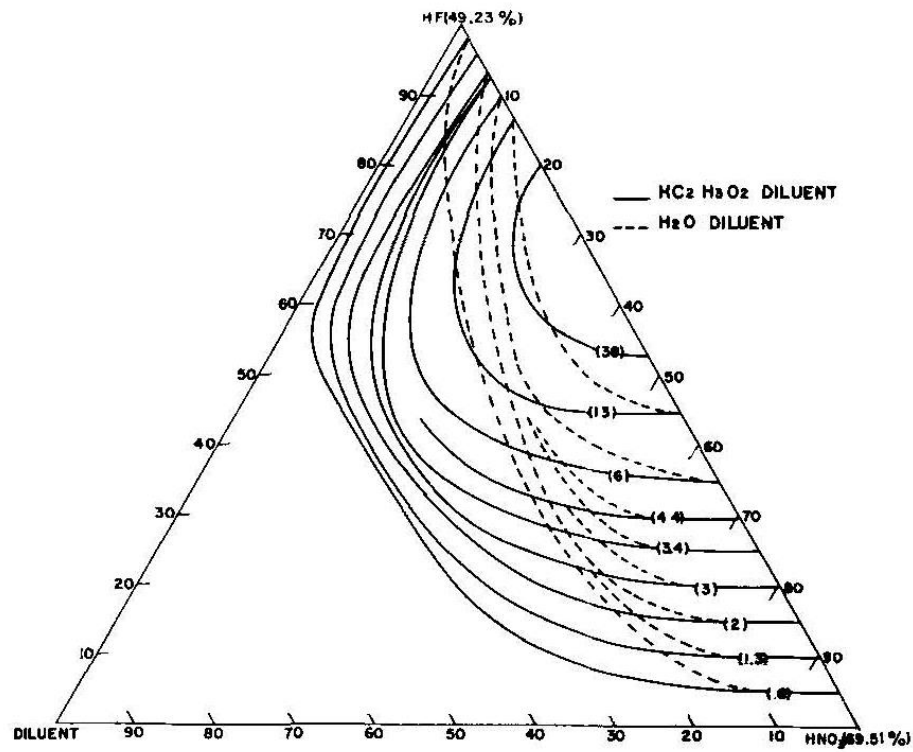
5. Etching 20

- | | | |
|-----|---|---|
| 5.1 | Sketch and explain two workflows for generating a metallic electrode structure (e.g., made from gold) | 8 |
| 5.2 | State the wet etching reaction of silicon in a mixture of nitric acid and hydrofluoric acid (HF) | 4 |
| 5.3 | Given the following etching diagram state three etching solutions which give an etching rate of 13 ($\mu\text{m}/\text{min}$) in acetic acid (solid lines) as diluent | 6 |



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5.4 Which one would you prefer and why?

2

